


# Profiling the Chinese Participants in Psychological Science: Non-WEIRD but Not Representative

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## Abstract

Psychological science aims at understanding the human mind and behavior, but it primarily relies on participants from Western, Educated, Industrialized, Rich, and Democratic regions, i.e., the WEIRD problem. This lack of diversity and representativeness of participants compromises the generalizability of psychological science. To address this issue, large-scale international collaborative projects have been initiated, and more data have been collected from non-WEIRD regions. However, it is unknown whether participants from non-WEIRD regions can represent their local population. In this preregistered report, we depicted the demographic profile of Chinese participants reported in 1, 000 empirical studies (n = 554, 794) published in five mainstream Chinese psychological journals and in 27 large-scale international collaborative projects (n = 29, 006, accounting for 2.12% of all participants). We found that age and sex were the most reported demographic variables, and other aspects of participants were less reported. The available demographic information revealed that Chinese participants were not representative of the population: most were young, well-educated, and resided in East China. Also, Chinese participants in large-scale international collaborative projects are less representative than those from Chinese journals in terms of geographical distributions. The demographic profile of Chinese participants in psychology is similar to their counterparts in many other countries and regions, suggesting the globalization of young and highly educated participants in psychological science. Together, our findings indicate that problems of representativeness are deeply entrenched in global psychological science and require coordinated cross-sector efforts.

**Keywords:** Meta-science; Population psychology; Representativeness; WEIRD; Generalizability

## 1 Introduction

Psychological science aims at understanding the human mind and behaviour. However, it largely relies on unrepresentative human samples: most human participants in published psychological studies are undergraduate students taking psychology courses from "Western, Educated, Industrialized, Rich, and Democratic" (WEIRD) regions (Henrich et al., 2010; Henry, 2008; Sears, 1986). For example, Arnett (2008) analyzed articles in six premier American Psychological Association (APA) journals and found that 96% relied on samples drawn from Western industrialized nations (Europe, North America, Australia, or Israel). More recent surveys have found little change in the past decade (Nielsen et al., 2017; Pollet & Saxton, 2019; Rad et al., 2018). However, the population in WEIRD regions is only consistent with less than 15% of the global population (United Nations, Department of Economic and Social Affairs, Population Division, 2024). The lack of representativeness in psychological science and related fields (such as cognitive neuroscience, Zuo et al., 2019) limits our understanding of the whole picture of the human mind and behaviour (Apicella et al., 2020; Barrett, 2020; Jones, 2010, White & Muthukrishna, 2026)[1] and may lead to incorrect policies (Arnett, 2008). This issue, combined with other methodological issues, has created a generalizability crisis in psychology (Yarkoni, 2022).

As a starting point to address this problem, researchers in the field started to include more diverse data. Many international collaborative projects have been initiated (Gordon et al., 2020; Moshontz et al., 2018). Typically, these projects invite collaborators globally, especially those from non-WEIRD regions, such as Asia, the Middle East, Latin America, and Africa. These efforts are commendable and have indeed increased the geographical diversity

[1] While it is generally accepted that samples should be representative of the target population, Mook (1983) argued that generalization may be misplaced in some cases where showing some effects do exist, even in rare and artificial settings, is valuable. This argument is invalid because most psychological research aims higher than mere existence of certain effects (e.g., guide the polices, IJzerman et al., 2020). Furthermore, focusing on a narrow subpopulation, we may miss phenomena that are outside that subpopulation and the consequences of these missed phenomena is unknown. Finally, the selection of samples reflects the fact that researchers themselves are from narrow subpopulation, they may prioritize the phenomena that are important to that sub-population and thus distort the overall picture of psychology.

and sample size of psychological science (Liu et al., 2025). These projects, however, have not examined whether data collected from non-WEIRD regions are representative of the local population. Leaving this issue unaddressed, these large-scale international collaborative projects may create an illusion that the representativeness problem can be solved by involving more researchers from non-WEIRD regions, ignoring the fact that data collected from non-WEIRD regions may also suffer from representativeness issues (see also Forscher et al., 2021). In fact, there are great variations within non-WEIRD regions (Ghai, 2021; Ghai et al., 2024).

To understand how representative the sample is in psychological research from a typical non-WEIRD region, China, we surveyed the studies conducted by Chinese psychological researchers. China is the world's second-largest economy and has the second-largest population, yet it has a very different history and cultural tradition from the West. In recent years, Chinese researchers have actively participated in international collaborative projects (e.g., Human Penguin Projects, Many Labs 5, Psychological Science Accelerator). However, it is unknown whether Chinese psychological participants represent the Chinese population. By the word "represent", we mean the sample in a study (or studies) should be a miniature of the targeted population without selection biases, or theoretically can be a miniature of the targeted population and without selection bias (Kruskal & Mosteller, 1979a, 1979c, 1979b, 1980; Kukull & Ganguli, 2012)

In the current study, we explored the representativeness of Chinese psychological participants to answer three questions (see Fig 1). Firstly, whether the characteristics of Chinese samples reported in large-scale international collaborative projects are similar to those reported in Chinese psychological journals. Secondly, to what extent the Chinese participants in psychological science represent the Chinese population, as compared with the census data from the National Bureau of Statistics of China and data from a large-scale social survey, the Chinese Family Panel Study (CFPS). Finally, we explored the shared and distinct patterns of Chinese samples and those from other countries and regions. In addition, we provided preliminary evidence about the "default participant" in Chinese psychological research. We compared the similarities and differences between the keywords of the large-

scale international collaborative projects and Chinese journals' articles based on bibliometric methods.

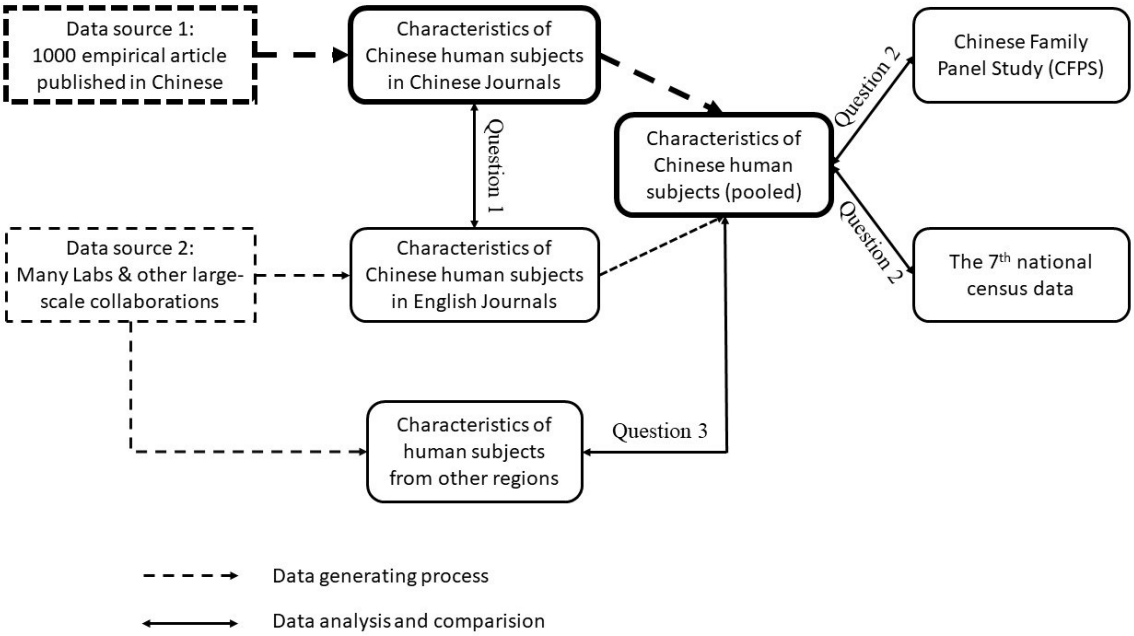



Fig 1. Schema of the current meta-research. Question 1: Whether characteristics of Chinese participants reported in large-scale international collaborative projects are similar to those reported in Chinese psychological journals, given they have the same target population; Question 2: To what extent the Chinese participants in psychological science can represent their target population, e.g., the whole population of census data from the National Bureau of Statistics of China and from a large-scale social survey, Chinese Family Panel Study (CFPS); Question3: What are the shared and distinct patterns of Chinese participants and participants from other countries and regions?

## 2 Method

### 2.1 Data sources

Data came from two sources. The first data source is 1, 000 empirical studies published in five mainstream Chinese journals: *Acta Psychological Sinica, Journal of Psychological Science, Chinese Journal of Clinical Psychology, Psychological Development and Education, Psychological and Behavioral Studies*. These journals are chosen for the following reasons. First, these five journals are indexed by CSSCI (Chinese Social Sciences Citation Index), which is regarded as an authoritative and comprehensive database for bibliometric studies of

China's social sciences (e.g., Gong & Cheng, 2022). Thus, all these five journals are selected as of high quality among all Chinese psychological journals. Second, these five journals cover most fields of psychology. Among them, *Acta Psychological Sinica*, *Journal of Psychological Science*, and *Psychological and Behavioral Studies* are comprehensive journals, studies from all sub-fields of psychology are included; *Psychological Development and Education* is the only journal for developmental and educational psychology in China; *Chinese Journal of Clinical Psychology* focuses on studies in clinical psychology and mental health. Of 1000 empirical papers, 500 were published between 2017 and 2018 that were selected by Wang et al. (2021). Another 500 papers were selected from the same five journals but published at different time periods. To be specific, we selected 250 articles published in 2008 and 250 articles in 2020~2021. The criteria and procedure of article selection are the same as Wang et al. (2021), and the special criteria and procedure can be found in the Supplementary materials (https://osf.io/y9hwq/files/vxapc).

The second data source came from large-scale international collaborative projects that are aimed at addressing the WEIRD problem. We primarily selected all Many Labs projects (Klein et al., 2018), Human Penguin Projects (Hu et al., 2019; IJzerman et al., 2018), and finished projects from PSA (Jones et al., 2021; Wang et al., 2021), the raw data of these projects are usually publicly available. We gathered a diverse array of other large-scale collaborations that contain samples from China by Dec. 2024. We reviewed the *Methods* and *Results* sections of these papers to determine the approach for acquiring data from these large-scale international collaborative projects. For papers that did not specify how to obtain the data or where the meaning of variables was ambiguous, we contacted authors via email. If no response was received after multiple rounds of email correspondence with the corresponding author or leading authors, the respective study was excluded from the present analysis.

Additionally, we exported the keywords of Chinese journals' articles and the large-scale international collaborative projects through CNKI (China National Knowledge Infrastructure) and Web of Science to compare the similarities and differences between them by using the bibliometric method.

**2.2 Articles code**

We manually coded the articles from the first source. The data extraction procedure consisted of three stages: pre-coding, coding, and proofreading.

In the pre-coding stage, we first developed the initial version of the coding manual based on the previous studies (e.g., Arnett, 2008; Nielsen et al., 2017; Pollet & Saxton, 2019; Rad et al., 2018). Then, at least two coders coded ten random articles independently, compared the results, resolved the differences, and revised the manual. After that, they coded another ten articles, compared the results, and revised the coding manual again. This procedure was iterated until the disagreement between the two coders was negligible.

When the formal coding manual was established, we started to code all 1000 papers. In this stage, we randomly divided the 1000 papers into several parts. For each part, two coders independently extracted data from papers based on the coding manual. Each coder went through the methods section and other parts to extract data. Note that studies used secondary data, data from web or app scraping, large-scale databases, using animals, case studies, or other inapplicable to our studies were excluded. For the remaining studies, we extracted basic information of articles (e.g., article title, first author, journal, study number, etc.), demographic information of participants (e.g., sex, age, socio-economic status, etc.), and other research-aiding information (e.g, target population, recruitment method, etc.). The detailed coding items can be found in "The Coding Manual V4" (https://osf.io/y9hwq/files/cyg43).

During the coding proofreading stage, the reliability of the final coding results was ensured through a combination of automatic text-match (see script "Coding_check.R", https://osf.io/y9hwq/files/3yzdb) and manual check. Specifically, the consistency of each item was automatically rated as one of three levels by matching the corresponding text: (1) complete inconsistency, which means the text by two coders has no overlap, (2) partial consistency, which means the text by two coders is partially overlapped, and (3) complete consistency, which means the text by two coders is exactly the same. This text-matching was then manually checked because the algorithm may not work well for Chinese characters. If an item was rated as inconsistent between coders (rated as 1 or 2), coders revisited the original text together to resolve discrepancies. Any remaining discrepancies, if present, were addressed via a final group consultation involving the project leader.

Notably, the participants' representativeness depends on the research question of a particular study. Therefore, the target population of our study was coded. However, because representativeness has not been taken seriously in the field (see Thalmayer et al., 2021), researchers may use vague terms for representativeness or generality. And based on feedback from coders in our pre-coding stages, it is difficult to code the target population. To improve the feasibility of coding the target population, we used a three-category coding schema: stated specific population; stated general population, and inferred general population[2]. To ensure coding transparency, we also included the excerpts from the original articles that were used for coding the target population.

### 2.3 Data analyses

We used R 4.1.1 (R Core Team, 2021) to preprocess, analyze, and visualize data. The code used for data analysis is available on OSF (https://osf.io/y9hwq/files/osfstorage) and GitHub (https://github.com/Chuan-Peng-Lab/chin-subj).

#### 2.3.1 Coding reliability and hypothesis testing

For coding reliability, we used the R package *irr* for calculating coding reliability (Gamer et al., 2019). And for hypothesis testing, we used Bayes factor analyses to conduct statistical tests on the data. The Bayes factor was chosen because it can provide evidence for both the null hypothesis and the alternative hypothesis (Dienes, 2016; Dienes & Mclatchie, 2018; Hu et al., 2018; Wagenmakers et al., 2018). Given that the final data are count data that are distributed in different categories of certain dimensions (e.g., sex, age bins), we used Bayesian multinomial test to test whether data from two sources differ from each other on certain dimensions (e.g., sex, age, educational attainment).

The multinomial distribution is a generalization of the binomial distribution to variables that can take values in $K \geq 2$ categories. The parameters of multinomial is a vector of probabilities, $\theta = (\theta_1, \theta_2, \ldots, \theta_K)$, with which *N* observations are distributed across *K*

[2] The stated specific population refers to the research conclusion is generalized to a specific group. Such as infants, adolescents, the elderly, etc. The stated general population refers to the situation where the researcher explicitly mentions that the research conclusion is applicable to "humans" or "Chinese people". The inferred general population refers to the article did not explicitly state its target population, and the coder makes the judgment based on the main idea of the research.

categories. The distribution of the parameters of multinomial distribution follows a Dirichlet distribution with concentration parameters $(\alpha_1, \alpha_2, \ldots, \alpha_K)$, where each $\alpha$ is larger than zero. In Bayesian multinomial test, the null hypothesis ($H_0$) is a point hypothesis that the parameters, $\theta_s$, of the observed percentage data equal to the expected, which is a point in the Dirichlet distribution, see Sarafoglou et al. (2023) for details. Usually, the $H_0$ is tested against the encompassing hypotheses, or alternative hypothesis ($H_a$), that all category proportions are free to vary. See the details of testing our first question below.

#### 2.3.1.1 Data analysis plan for the first question

Our first question is whether there are similarity between Chinese human subjects reported in Chinese journals and in large-scale international collaborative projects. Given that studies from Chinese psychological journals may have different target populations as compared to international collaborative projects, we compared samples from studies that share the same target population. More specifically, only if articles from Chinese psychological journals and large-scale international collaborative projects targeted the general population (inferred or stated Chinese population or humans), we compared their sample characteristics. In the same vein, samples from other shared target populations by both Chinese psychological journal articles and large-scale international collaborative projects, e.g., adolescents, also were compared.

We first visualized the proportion of the reported information of subjects in several dimensions (e.g., sex ratio). These visualizations provided intuitive information.

Then, we compared subjects from Chinese psychology journals and Chinese subjects from large-scale international collaborative projects with regard to sex, age, educational attainment, and, if possible, geographical distribution. As mentioned above, we used Bayes multinomial test for our statistical inference. Here we elaborated on how to compared the two data in terms of sex ratio and age. We implemented Bayesian multinomial test using R code based on JASP 0.16.4 (see the R script in R Notebook for more details, https://github.com/yuelei911/chin-subj/tree/master/3_Data_Analysis).

***Sex ratio*** We tested whether the sex ratio of subjects from international collaborative projects (observed data) was sampled from a distribution with parameters equal to the

proportions of data extracted from Chinese psychology journals (expected data). The $H_0$ and $H_a$ are specified as below:

$H_0$: the observed data are sampled from a multinomial distribution with a parameter $P = Pr(x_1, x_2 \mid n = N, p_1, p_2)$, where $(p_1, p_2)$ equals expected data, which is a fixed point in a *Dirichlet* distribution.

$H_a$: the observed data are sampled from a binomial distribution with parameter $P = Pr(x_1, x_2 \mid n = N, p_1, p_2)$, where $(p_1, p_2)$ is distributed as a Dirichlet distribution with concentration parameter $(\alpha_1, \alpha_2)$.

We used the Bayesian multinomial test for hypothesis testing. The prior is non-informative prior, i.e., *Dirichlet*. Bayes factor was interpreted as recommended in (Wagenmakers et al., 2018): $BF_{a0} >= 10$ or $\ln(BF_{a0}) >= 2.303$ means strong evidence for $H_a$, and $6 <= BF_{a0} < 10$ or $1.792 <= \ln(BF_{a0}) < 2.303$ means moderate evidence for $H_a$, $BF_{a0} <= 1/10$ or $\ln(BF_{a0}) <= -2.303$ means strong evidence for $H_0$, and $1/10 <= BF_{a0} < 1/6$ or $-2.303 <= \ln(BF_{a0}) < -1.792$ means moderate evidence for $H_0$.

Based on the simulation in Stage 1 (see S2 in Supplementary materials, sample size should be greater or equal to 1200), we used the non-informative prior because the final data have a large enough sample size (the 27 large-scale collaborative projects included 29, 006 Chinese participants and the 1000 include 554, 794 Chinese participants). Thus, the sample in our study is enough for our Bayesian multinomial test for sex ratio.

***Age distribution*** For the age distribution, we created age bins using two different approaches. The first approach was based on the developmental stage, which was more important in psychological research than biological age itself. We created five age bins based on developmental stages: 0 ~ 17 (children and adolescents), 18 ~ 25 (early adulthood), 26 ~ 40 (middle adulthood), 41 ~ 59 (later adulthood), and >= 60 (elders). The second approach was similar to census data's age bins: 0 ~9, 10 ~ 19, 20 ~ 29, 30 ~ 39, 40 ~ 49, 50 ~ 59, and >=60[3]. Unless stated, we reported statistical results based on the first approach of age bins

[3] Note that a more fine-grained age bins (with 5-year intervals) were only used for pyramid plot but not for statistical analysis. The four age bins were adopted for the third research question to better reveal data differences.

in the main text and results based on the second approach in the supplementary results. In this case, the $H_0$ and $H_a$ are specified as below:

$H_0$: the count data in different age bins of Chinese participants from large-scale international collaborative projects are sampled from a multinomial distribution with parameters $P = Pr(x_1, x_2, \ldots, x_5 \mid n = N, p_1, p_2, \ldots, p_5)$, where $(p_1, p_2, \ldots, p_5)$ is the percentage data of Chinese subjects from Chinese psychology journals, which can be view as a point in a *Dirichlet* distribution.

$H_a$: the count data in different age bins of Chinese participants from large-scale international collaborative projects are sampled from a multinomial distribution with parameters $P = Pr(x_1, x_2, \ldots, x_5 \mid n = N, p_1, p_2, \ldots, p_5)$, where $(p_1, p_2, \ldots, p_5)$ is distributed as a *Dirichlet* distribution with concentration parameter $(\alpha_1, \alpha_2, \ldots, \alpha_5)$.

As for the sex ratio, we also used the Bayesian multinomial test for hypothesis testing, with non-informative prior, i.e., *Dirichlet* distribution. The resulting Bayes factor was interpreted same as the sex ratio.

Our simulation in Stage 1 revealed that with a sample size of 1200 and non-informative prior, Bayesian multinomial test can detect a mean difference of 0.95 years (in the best case) to 1.6 years (in the worst case) when participants were young or mid-aged adults (see S3 in Supplementary materials). As the final sample size was larger than 1200, the current setting was sensitive to detect the even smaller difference.

**2.3.1.2 Data analysis plan for the second question**

The second question of this study is whether all Chinese sample data available, regardless of the sources of the data, come from a very narrow slice of the Chinese population. Given sample representativeness indeed depend on the target population, we further distinguished two types of analyses. For studies that targeted the general population, inferred or stated, we compared their sample characteristics to the whole census data from the National Bureau of Statistics.

For studies that targeted a specific population, we compared the sample characteristics to that specific population selected from census data. When the information of that specific population is not available in census data, we searched for other reliable data sources as the reference data. We also used the Bayesian multinomial test for statistical inference. For those

targeted at the general population, we used the pooled Chinese subjects' data from both sources as the "observed" and the census data as the "expected". The age bins were the same as we tested the first hypothesis. The prior setting of Bayes factor analyses and the criteria for interpreting the Bayes factor were the same as in testing our first question. Note that the sample size might be smaller than 1200 for studies aimed at generalized to a specific population. In this case, we interpreted our results as exploratory.

#### 2.3.1.3 Data analysis plan for the third question

For the third question, whether Chinese psychological samples differ from other countries' samples, we visualized the distributions of the participants from different countries and visually compared other countries with the Chinese samples. We also compared each country's data with Chinese data using the Bayesian multinomial test, where the other countries' data were treated as "observed" and Chinese data as "expected. The null hypothesis ($H_0$) is that the parameters of the observed sex and age distributions equal the expected proportions (i.e., Chinese psychological samples). The alternative ($H_a$) is that those parameters are free to vary. The prior setting of Bayes factor analyses and the criteria for interpreting the Bayes factor were the same as in testing our first question. Note that the sample size might be smaller than 1200 for some countries, in these cases, we interpreted our results as exploratory.

### 2.3.2 Exploratory analyses

Previous studies found that whether mentioning the demographic information in the article's title influences research evaluation (Kahalon et al., 2022), researchers may omit participants' information to reduce the bias of research evaluation in peer review. Meanwhile, many researchers found psychological studies dependent on university students (Arnett, 2008; Sears, 1986), authors might assume university students are "default participants" and then choose not to report this sample type or other demographic information in their papers. Therefore, we analyzed the reporting of participant information in the abstracts. First, we examined whether authors mentioned "university students" in the abstracts when the studies included university student samples. Second, we analyzed the reporting of various sample

types of participant information across different participant groups. The results are available at the supplemental materials.

Furthermore, we conducted the following exploratory analyses: (1) We described the temporal changes in demographics reported in the Chinese papers across different years; (2) When testing the representativeness of Chinese participants, we separately compared participants from the Chinese journals and international collaborative projects against Census data and CFPS data; (3) We examined whether the profile of participants from different study types differed; (4) We performed bibliometric analyses to compare the similarities and differences of the keywords across the two data sources. All relevant detailed results are available in the supplemental materials.

### 2.4 Disclosure

We submitted this project as the registered report of PCI and the current document is at Stage 2 of the registration. The in-principle accepted the Stage 1 protocol is available in https://osf.io/ehw54.

#### 2.4.1 Deviation from the Stage1 protocol

We used the Preregistration Deviations Table created by Willroth and Atherton (2024) to document all deviations from Stage 1 protocol. All these deviations did not substantially affect the strength of the evidence, and the Preregistration Deviations Table is available at the Table S2.

## 3 Results

### 3.1 Coding reliability

For the first data source, the interrater agreement was calculated using *irr* package in R for Krippendorff's α (Gamer et al., 2019), which ranged from 0.76 to 0.99. All categorical variables were also estimated using the agreement rate in our study[4], which ranged from 87% to 99%. As mentioned above (Section 2.2 Articles code), discrepancies were found by text-matching and manual reviews and resolved through discussion among coders. Similarly, we

[4] For articles published in Chinese psychological journals, we computed the following coding items: abstract, target population, age, sex, educational attainment, sampling method, subjects recruitment method, sample size, and study type. These items were used in the analysis presented in this paper.

applied the aforementioned methodology to assess the coding reliability of large-scale international collaborative projects. Since this dataset only coded the target population, the reliability for this item was extremely high, with both Krippendorff's α and the agreement rate equal to 1.0.

### 3.2 Overview of participants

#### 3.2.1 The demographics reported in the Chinese papers

The 1, 000 Chinese papers included 1, 264 studies and 554, 794 participants. Among the 1000 articles, 622 targeted the special population, while 378 targeted the general population (see supplementary document *S5* for more detailed information). Regarding the demographic information reported in studies, we found sex and age were most frequently reported (Fig 2), with 90.59% of studies reporting sex and 81.72% reporting age, respectively. For the other demographic information, 63.13% of studies reported recruitment method, 56.25% reported recruitment area, 45.75% reported educational attainment, 34.31% reported occupation, 33.89% reported sampling method, and 28.23% provided information on socioeconomic status. Ethnicity (3.01%) and religious beliefs (0.16%) were scarcely reported. Notably, our exploratory analysis revealed that virtually no consistent increase in demographic information reporting across different time periods except for age, sex, and sampling methods (see the Fig S3). We also visualized the demographics reported for the two target populations (Fig S4). In addition, we observed non-standard practice in the reporting of demographic information. For instance, age reported was ambiguous in 134 studies, where only age ranges were provided. This issue also applied to socioeconomic status: of the 341 studies reporting socioeconomic status related information (e.g., occupation, educational attainment, income, subjective social class), 251 used different single indicators to measure socioeconomic status, while the remaining studies used different composite indicators.

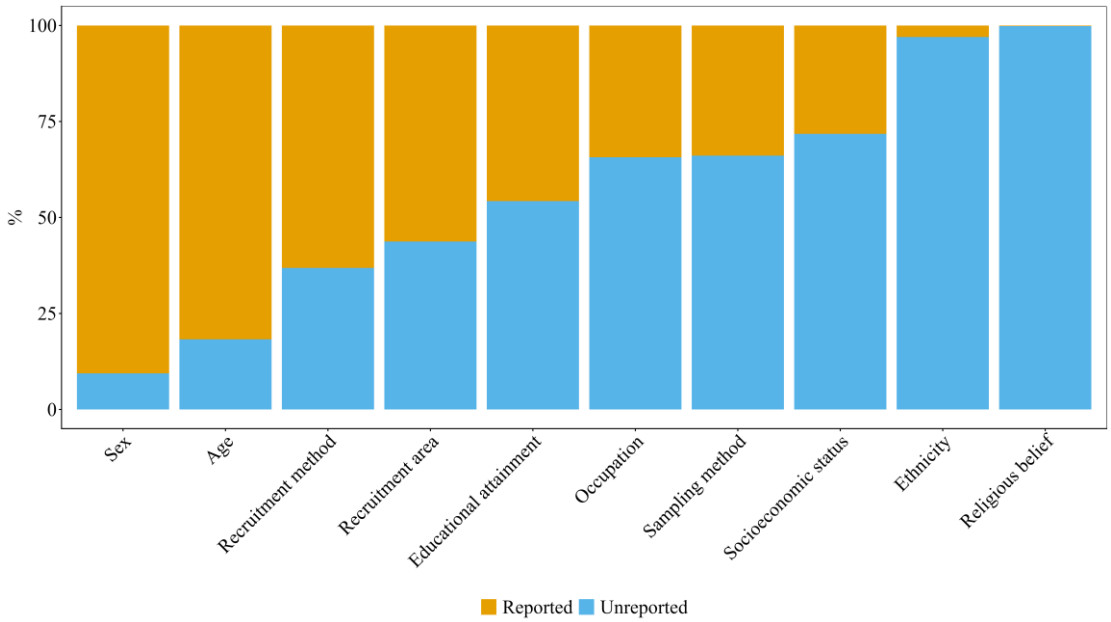



Fig 2. Proportion of participants with demographic information reported in Chinese papers

### 3.2.2 The demographics reported in large-scale international collaborative projects

We obtained data from 27 unique large-scale international collaborative projects from 31 papers (all projects are available in "BTS_projects.xlsx", https://osf.io/y9hwq/files/azemd[5]), which included data from 218 countries and regions, involving a total of 29, 006 Chinese participants (accounting for 2.12% of all participants in large-scale international collaborative projects, all countries and regions of participants' information can be available in "BTS_participants.xlsx", https://osf.io/y9hwq/files/c6tmu). Similar to the pattern in Chinese journal papers, there were notably differences in the reported demographic data across different projects. Specifically, among the 27 projects, all provided information regarding sex and country/region. Twenty-six of them contained data on age, and seventeen provided information related to education. Regarding the uniformity of demographic reporting, for instance, for the participants' geographic background information, 24 projects directly reported "country" information, 21 provided "region" information, while some others provided relevant details such as "university", "country residence", "country birth", "province/city", and "citizenship/nationality".

[5] "BTS" in our study refers to "big-team science", we used it interchangeable with "large-scale international collaborative projects"

Therefore, in the subsequent analysis, only the participants' sex, age, educational attainment, and recruitment region were considered.

### 3.3 Comparing Chinese papers and large-scale international collaborative projects

To test our first hypothesis, we compared Chinese participants from papers published in Chinese journals and from large-scale international collaborative projects using Bayesian multinomial test. Our first research hypothesis was that the demographic distributions were identical across these two Chinese samples. As in our stage 1 plan, we first categorized papers/projects into two groups: those aiming to generalize their conclusion to the general population and those only aiming at infants, and only compared those that shared the same target population. Note that only the 378 articles that targeted the general population and the 37 articles that targeted infants were used for the comparison below, the rest of the articles targeted other populations and did not match the target population of large-scale international collaborative projects (see supplementary results S5 in the supplementary materials for numbers of articles for each target population).

For the sex ratio of participants, available from 341 Chinese journal papers and 26 large-scale international collaborative projects that targeted the general population. We found moderate evidence supporting $H_0$, $\ln(BF_{a0}) = -1.64$ (see Fig 3A). For papers/projects that targeted infants, we compared data from Lucca et al. (2025) to data from 37 Chinese journal papers that also targeted infants and found moderate evidence supporting $H_0$, with $\ln(BF_{a0}) = -1.50$.

For the age distribution, we obtained information from 208 Chinese papers and 25 large-scale international collaborative projects that provided detailed age information and targeted the general population. We found extreme evidence for $H_a$: samples from Chinese papers were different from those from large-scale international collaborative projects, $\ln(BF_{a0}) = 28.06$ (Fig 3B). While both samples peaked at young age (between 20 and 24 years old), samples from Chinese journals were more concentrated than those from large-scale international collaborative projects. For projects targeting the special subpopulation, we identified only one large-scale international collaboration project, Lucca et al. (2025), focused on infants. Consequently, we did not compare age data between the two data sources.

In terms of distribution of educational attainment, as many papers did not report the exact information, we used two different educational attainment as cutoffs to categorize participants included in this analysis: college or high school. There were 339 Chinese papers and 16 large-scale international collaborative projects that provided educational information and targeted the general population[6]. When using college level education as the cutoff, we found moderate evidence supporting $H_0$ (College-or-above vs less-than-College: $\ln(BF_{a0}) = -1.99$, see Fig 3C). When using high school level as the cutoff, we also found moderate evidence supporting $H_0$ (high-school-or-above vs less-than-high-school: $\ln(BF_{a0}) = -1.17$, see Fig 3D). These results suggest that participants in large-scale international collaborative projects were comparable to those in papers published in Chinese journals.

For regional distribution, we obtained province-level geographical data from 130 papers and 6 large-scale international collaborative projects (see Fig 4A and Fig 4B). We used two different approaches to cluster provinces to several geographical regions in China (see Fig 4C and Fig 4D) when conducting statistical analyzes. When using the four-region classification (Central China, East China, West China, Northeast China), we found extremely strong differences in samples between Chinese papers and large-scale international collaborative projects ($\ln(BF_{a0}) = 33.49$). Under the alternative seven-region approach (East China, Central China, South China, Southwest China, Northwest China, Northeast China), the Bayesian multinomial test also uncovered extreme evidence of regional imbalance between samples of Chinese papers and large-scale international collaborative projects ($\ln(BF_{a0}) = 50.64$). These results suggested that Chinese participants are less representative of the Chinese population in terms of geographical distribution. For projects targeting the special subpopulation, the comparison is not feasible because the province level data is unidentifiable for mainland China in the only one large-scale international collaboration project, Lucca et al. (2025).

[6] Thirty-one articles adopted young adults as the study sample, while the rest used student samples. For 31 Chinese papers, because some articles had incomplete education information, only 22 articles in the college category and 15 articles in the high school category were included in our analysis. For large-scale international collaboration projects, one of the 17 projects was removed because its education information could not be converted into the categories used for later analysis.

In summary, samples from large-scale international collaborative projects and Chinese journals differed in age and geographical distribution but not in sex and educational attainment. Thus, our first research hypothesis was partially supported.

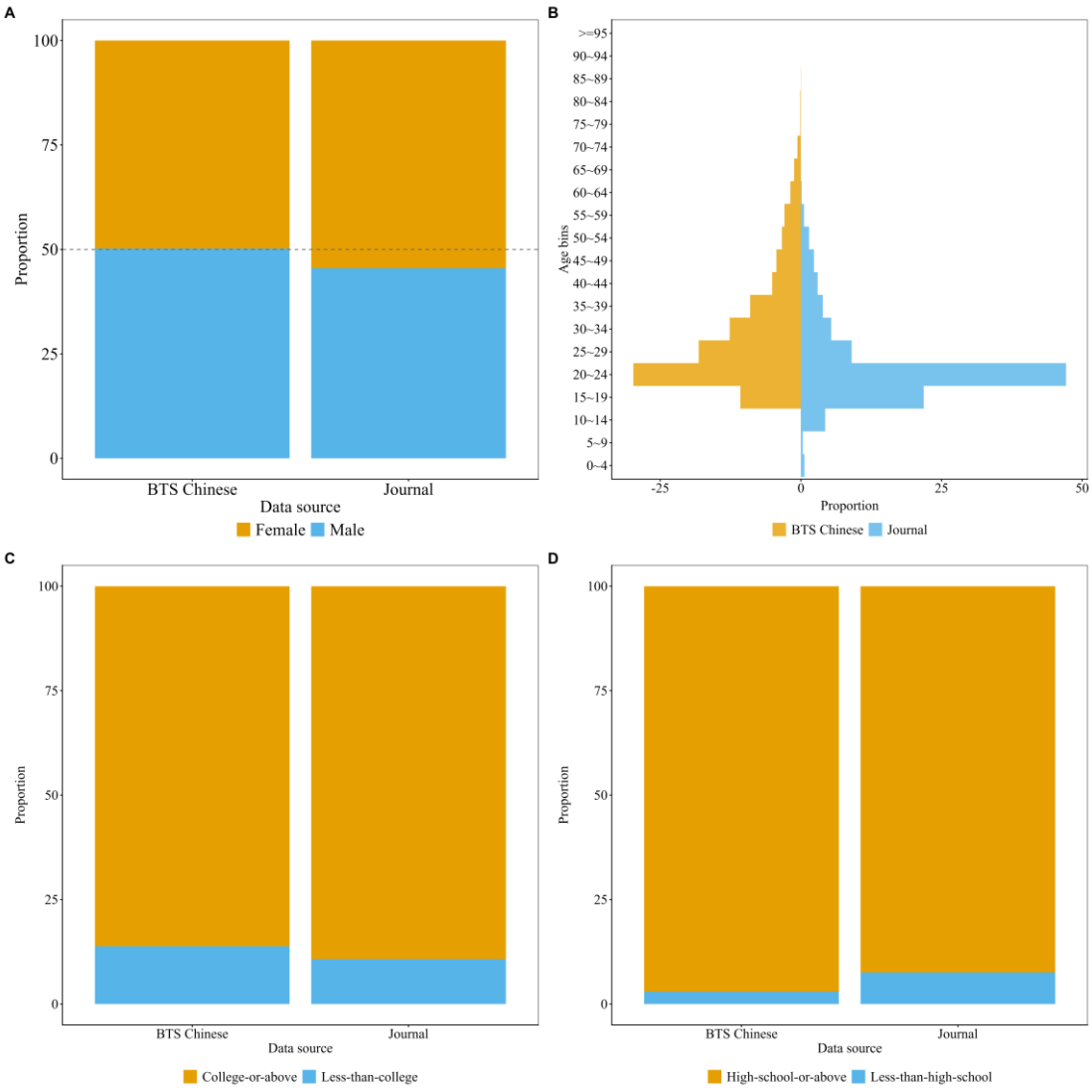



Fig 3. sex, age, and educational attainment distribution of Chinese participants from large-scale international collaborative projects and Chinese journal papers. BTS Chinese: the Chinese participants from large-scale international collaborative projects; Journal: the Chinese participants from Chinese journal papers

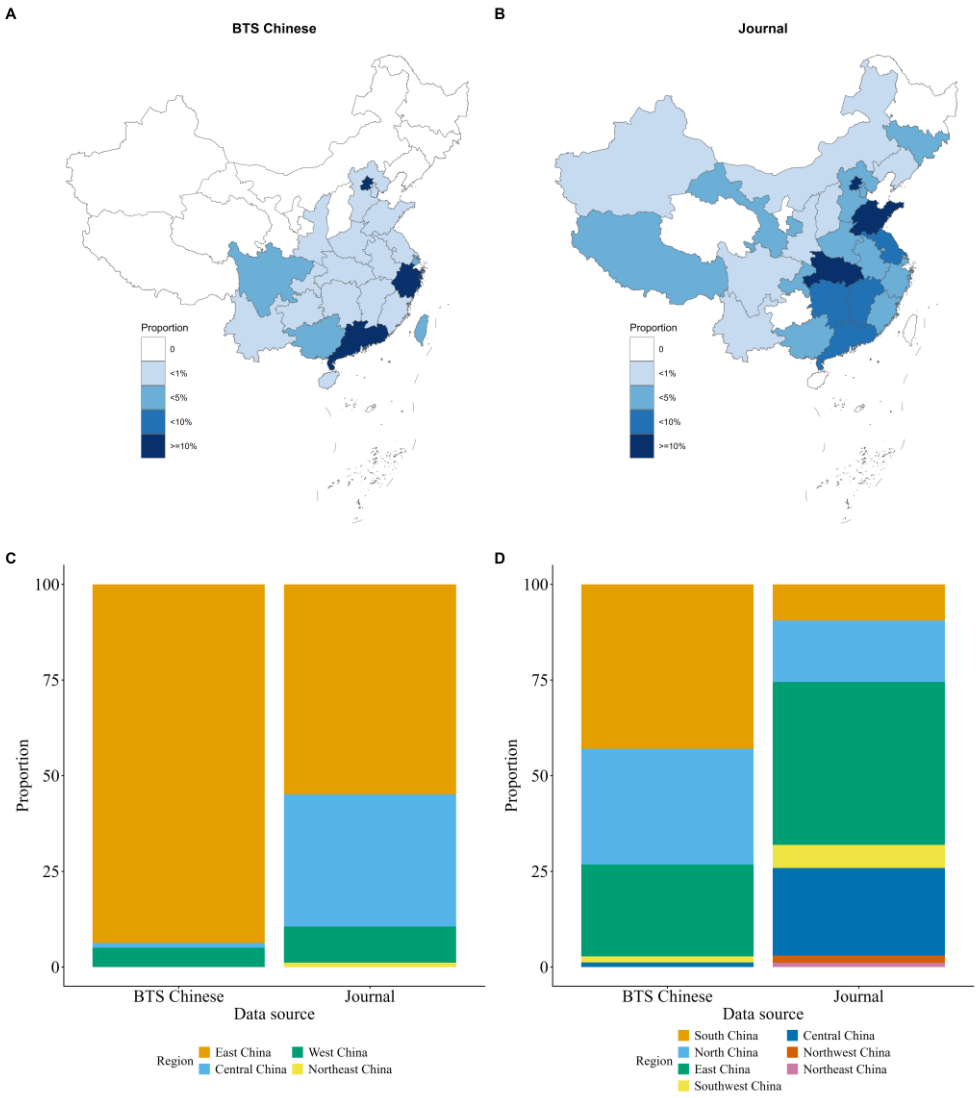



Fig 4. The regional distribution of Chinese participants from Chinese journal papers and large-scale international collaborative projects.

### 3.4 Comparing Chinese samples and samples from census data and CFPS

To assess whether Chinese participants in psychological research are representative of China's population, we compared their sex, age, educational attainment, and recruitment regions against census data and CFPS data. Our second research hypothesis was that the Chinese samples come from a very narrow slice of the Chinese population. Specifically, Our $H_a$ posited that the demographic distributions of samples in psychological research differed from those of the Census/CFPS data. Although we found evidence that the samples from Chinese papers and large-scale international collaborative projects were different on some

demographic dimensions, we reported the results of pooled data in the main text and presented the separate results in the supplemental materials.

For the sex ratio in studies that targeted the general population, our results revealed moderate evidence supporting $H_0$ and indicated there was no difference between Chinese participants in psychological studies and the 7th Census ($\ln(BF_{a0}) = -1.64$) or CFPS 2018 ($\ln(BF_{a0}) = -1.84$), see Fig 5A. We further examined the participants in studies that targeted specific subpopulations. For those targeting college students, we compared the sex ratio of Chinese participants from these studies with the statistical monitoring report on the *Outline for the Development of Chinese Women (2021–2030)*, released by the National Bureau of Statistics of China in 2022, and found moderate evidence in favor of $H_0$ ($\ln(BF_{a0}) = -1.67$). For those studies that targeted preschool children, we compared the data from Chinese participants from these studies with the preschool children subpopulation in the 7th Census data, and the results showed the same pattern ($\ln(BF_{a0}) = -2.06$).

For age distribution, we could only compare the general target population between Chinese participants in psychological research and the 7th Census data. Our results revealed extreme evidence in favor of $H_a$, and the differences were observed both in the figure and the Bayesian multinomial test ($\ln(BF_{a0}) = 85.84$).

For the educational attainment ratio in studies that targeted the general population, we also compared the educational attainment of Chinese participants in psychological research with Census data and CFPS 2018, respectively, using two different cutoffs. When using a college-level cutoff, we found extreme evidence that the Chinese participants in psychological research differ from the Census data (college: $\ln(BF_{a0}) = 110.03$). Similarly, when using high school level as the cutoff, extremely different was observed between the two data sources ($\ln(BF_{a0}) = 72.11$). Note that the Chinese participants in psychological research were also different from those in the CFPS 2018 for both educational attainment cutoffs (college: $\ln(BF_{a0}) = 64.66$; high school: $\ln(BF_{a0}) = 141.73$), see Fig 5C and 5D.

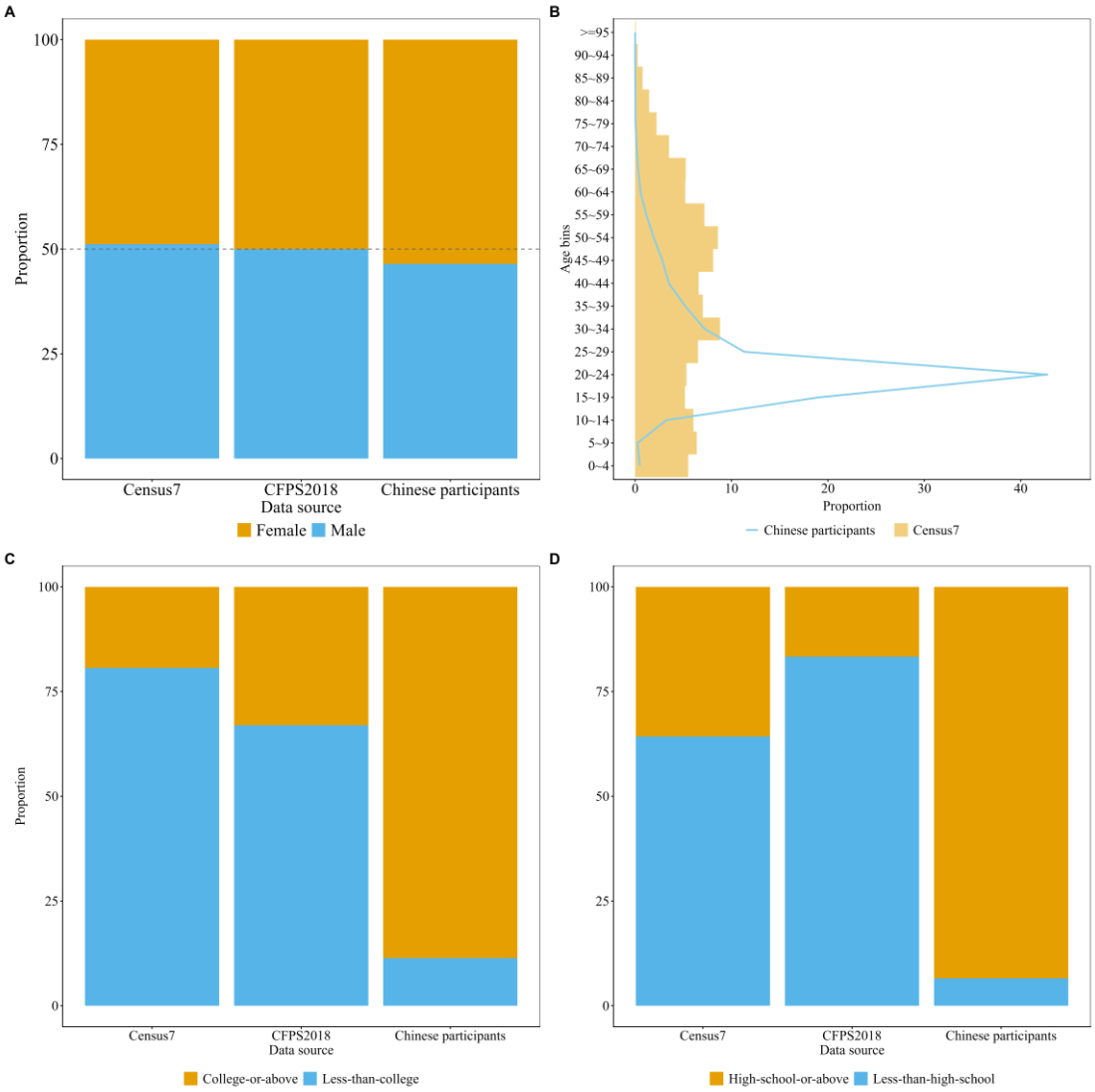



Fig 5. sex, age, and educational attainment distribution of Chinese pooled participants. Census7: the 7th National Population Census of China; CFPS2018: China Family Panel Studies 2018; Chinese participants: the participants in psychological research.

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Additionally, we also analyzed the regional distribution between participants' from the Chinese psychological participants, the 7th Census, and the CFPS 2018. The geographical distribution figure is shown in Fig S11. We still used two different Chinese major regional classifications to further visualize the regional data (see Fig 6) and analyzed the differences. When using the four-region classification, we found extremely strong differences in samples between Chinese psychological studies and the 7th Census ($\ln(BF_{a0}) = 9.35$), as well as between Chinese studies and CFPS 2018 ($\ln(BF_{a0}) = 20.73$). Under the alternative classification, the Bayesian multinomial test uncovered very strong/extreme evidence of

regional imbalance between samples of Chinese psychological studies and both the 7th Census data ($\ln(BF_{a0}) = 3.63$) and CFPS 2018($\ln(BF_{a0}) = 21.42$).

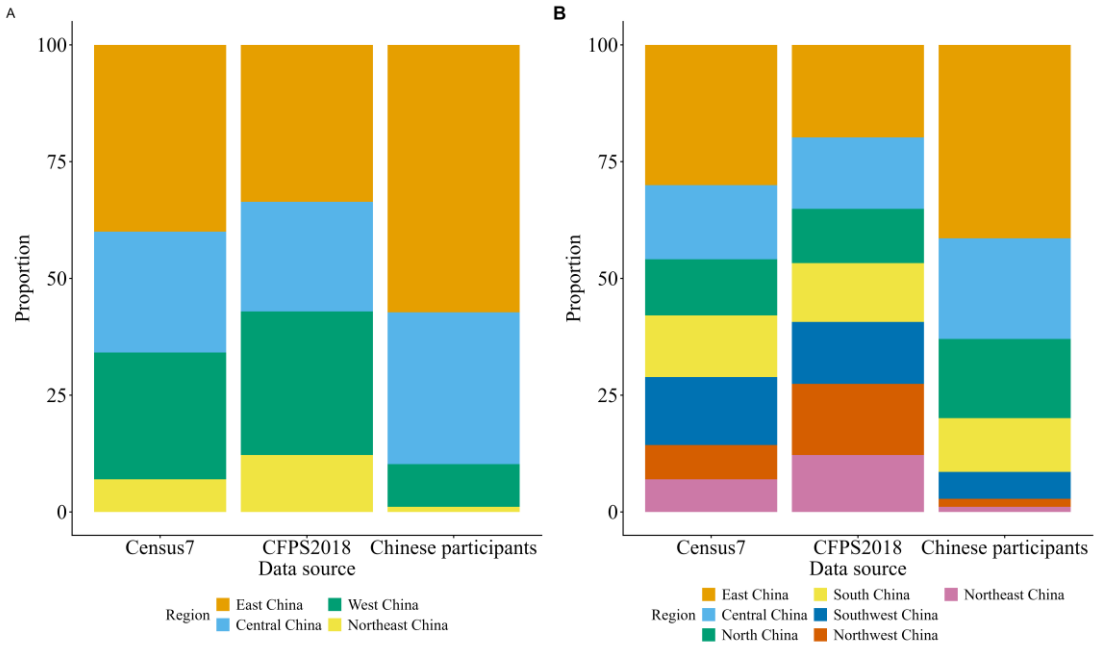



Fig 6. Major regional distribution in China: 7th Census, CFPS 2018, and Chinese participants in psychological research.

Overall, compared with Census/CFPS data, we found that Chinese participants are younger, highly educated, and mostly from East China. Therefore, Our second research hypothesis was supported.

### 3.5 Comparing Chinese participants and participants from other countries and regions

Our final analysis explored the similarities and differences in patterns between Chinese participants in psychological research and participants from other countries and regions. For the sex ratio that targeted the general population, the pairwise Bayesian multinomial test revealed that data from 27 countries and regions (among a total of 129 countries and regions) had the same sex ratios compared to Chinese psychological participants (see Fig 7A and 7B). We also compared Chinese participants and participants from other countries and regions within the special population. We analyzed Lucca et al. (2025), which aims to generalize their conclusion to infants. The pairwise Bayesian multinomial test revealed that data from 4 countries (among a total of 9 countries and regions) had same sex ratios compared to Chinese psychological participants (see Fig 8A and 8B). Suggesting the balanced pattern of Chinese

participants is not common in the participating countries and regions. Thirty-nine countries and regions showed different sex patterns from those of Chinese participants in the general population. The participants from these countries and regions showed a considerably higher proportion of female participants (2 countries and regions) or a considerably higher proportion of male participants (37 countries and regions). This indicate an imbalanced sex ratio in these countries and regions.

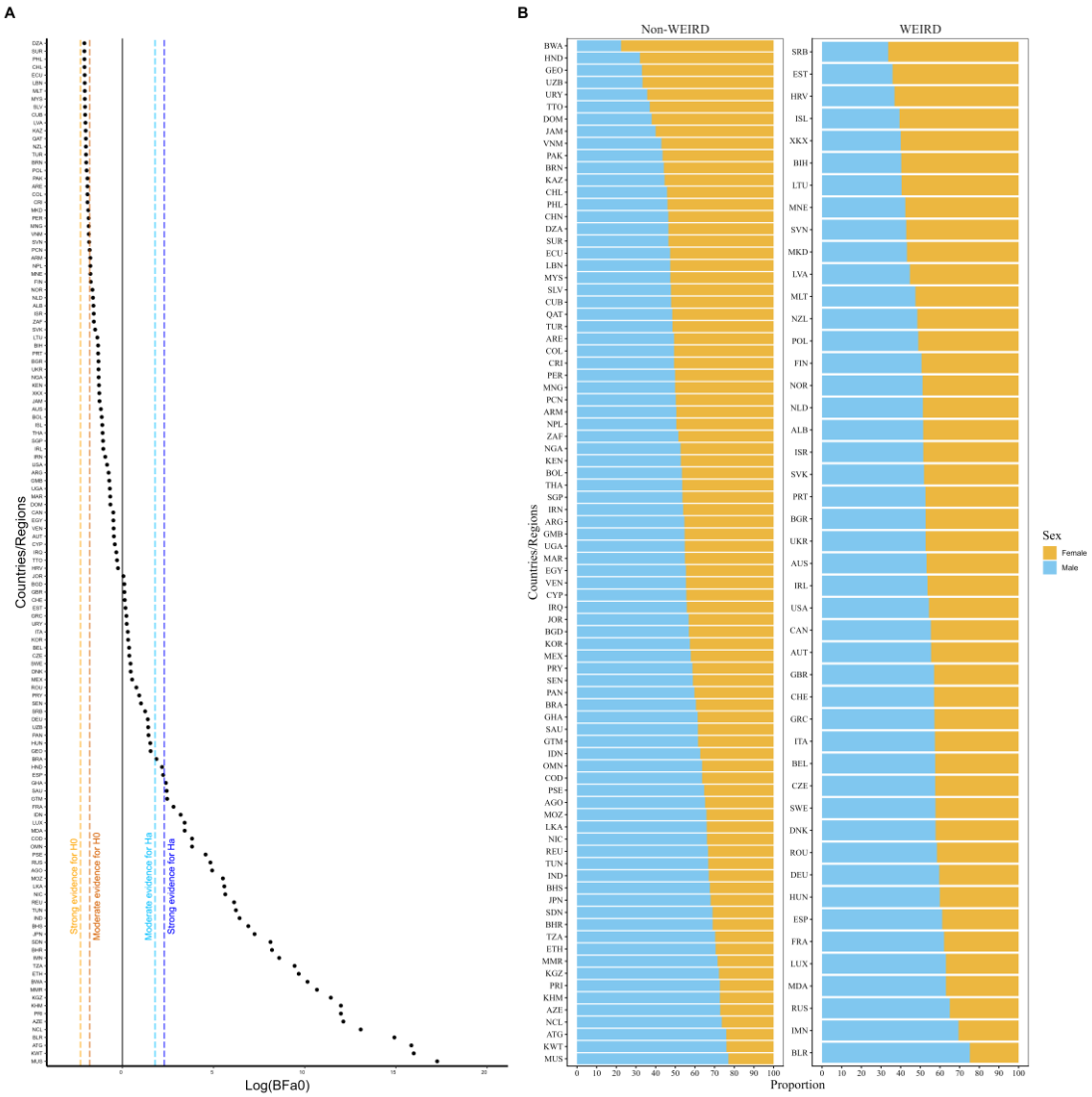



Fig 7. The sex distribution for psychological participants. (A) Pairwise comparisons of sex ratio between Chinese psychological participants and available data from other countries and regions; (B) Sex ratio of all data. Country code is ISO3.

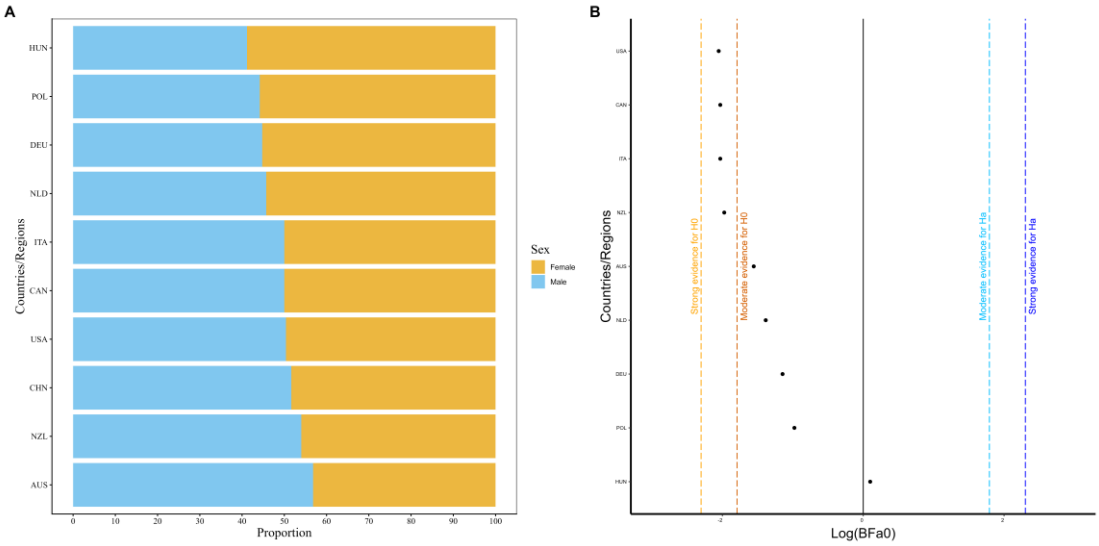



Fig 8. The sex distribution for psychological participants in Lucca et al. (2025). (A) Sex ratio of all data; (B) Pairwise comparisons of sex ratio between Chinese psychological participants and available data from other countries and regions. Country code is ISO3.

For the age distribution that targeted the general population, the pairwise Bayesian multinomial test revealed moderate evidence that participants from 5 countries and regions (among a total of 127 countries and regions) were the same as Chinese psychological participants (see Fig S7). Twenty-three countries and regions had a proportion of over 60% for participants in 18-25 age bracket, with the proportion of the Chinese participants in this age group reaching 59.73%. If we pool the 18-25 and 25-40 together as the early adulthood, then we found that 124 out of 127 countries and regions had more than 60% participants in this age group (see Fig 9). These results indicate that the psychological participants from most countries and regions are similar: most participants are young adults.

Based on the above findings, our assumption that more than half of other countries and regions share characteristics with China was only confirmed regarding age.

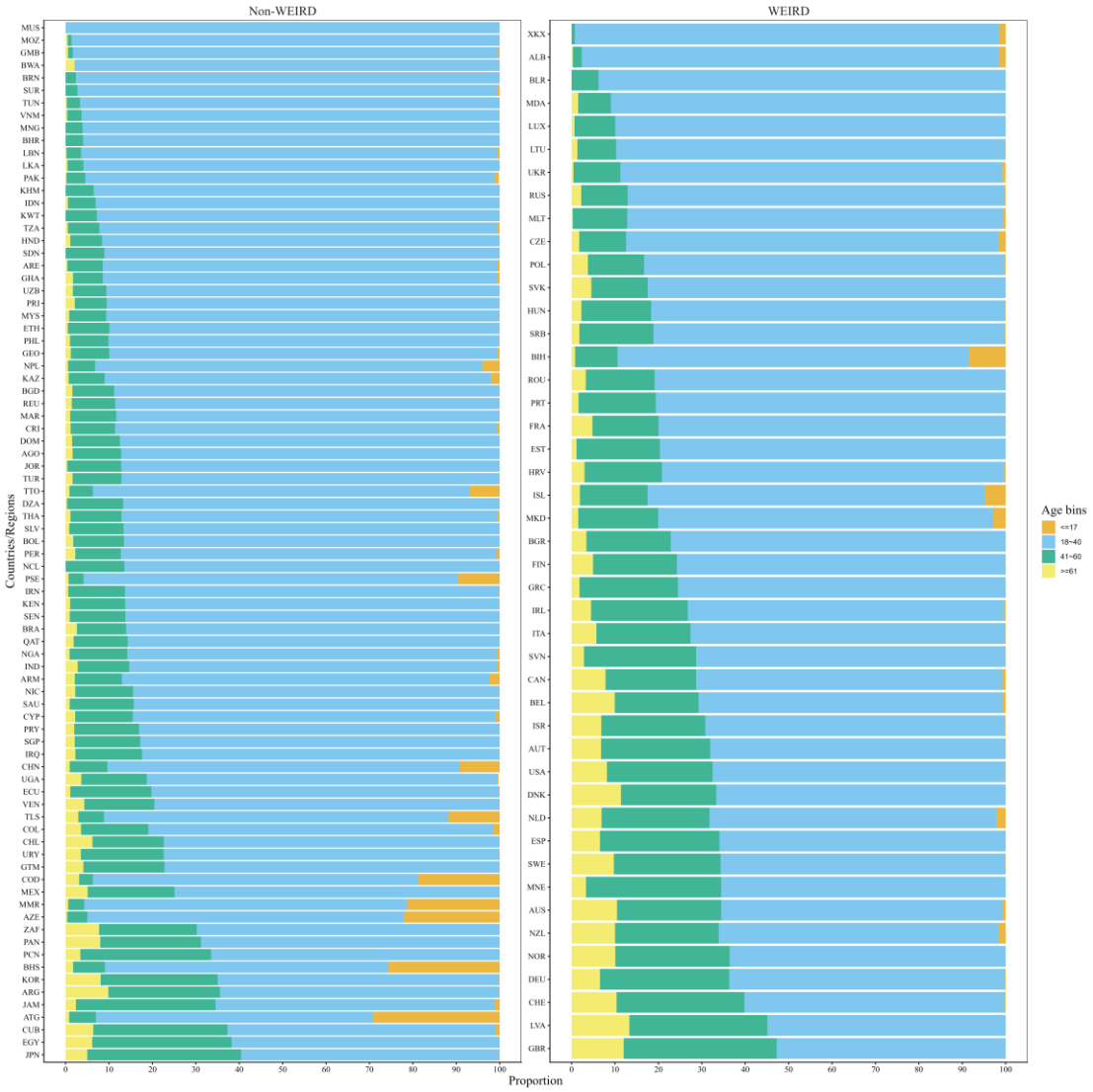



Fig 9. The age distribution from psychological participants with 4 developmental stages. Country code is ISO3.

## 4 Discussion

In this registered report, we analyzed the demographics of Chinese participants from 1,000 randomly selected papers in five top Chinese psychology journals and 27 unique large-scale international collaborative projects from 31 papers. We found that sex and age were the most frequently reported demographic information, and socioeconomic status, religion, and ethnicity were less frequently reported. The demographics of Chinese participants from Chinese psychological journals and large-scale international collaborative projects were similar in sex ratio and educational attainment but different in age distribution and geographical distributions. Also, we found that the age distribution of Chinese participants

was similar to that of other countries and regions. Unfortunately, more nuance within- or cross-country comparisons are not possible because of the lack of information.

Our findings revealed the similarity and difference of demographic patterns between Chinese psychological journals and large-scale international collaborative projects. Collectively, both two data sources indicated that Chinese participants were likely to be young, highly educated individuals, and resided in East China. As in the international journals, college students are likely to be the default participants in Chinese journals (Arnett, 2008; Nzinga et al., 2018; Sears, 1986). However, participants recruited by large-scale international collaborative projects are more concentrated in East and Central China. This may be attributed to the geographic locations of Chinese researchers who participated in large-scale international collaborative projects. Given that participating in these projects requires proficiency in English and richer research resources, this pattern is not surprising because more elite psychological departments are in East and Central China (e.g., as reflected in QS World University Rankings by Subject for Psychology). Additionally, participants from Chinese psychological journals are more concentrated in adolescents and young adults, with the proportions of the 15–24 age group being at least 1.5 times that of large-scale international collaborative projects. One possible reasons is that large-scale international collaborative projects may encourage researcher to diversify the age range of participants. This explicit recommendation may be implemented through online data collection platforms by restricting the age distribution of respondents (Del Ponte et al., 2024).

Our results provide the first empirical demographic profile of Chinese participants in psychological science. While Chinese participants are generally balanced in sex ratio, the age distribution, educational attainment, and geographical distributions are not representative of the country's population. These results suggest that Chinese researchers may be aware of the sex balance when recruiting participants but pay less attention to other demographics, resulting in samples that hardly represent the Chinese population. More importantly, the pattern of the Chinese participants' demographics is largely shared across different countries and regions (e.g., Ghai et al., 2024). While the sex ratio shows large variability across countries and regions, the age distribution pattern is similar: participants from most countries and regions are packed at the 18-40 years old. These findings suggest that the over-reliance on

young and, potentially highly educated, participants in psychological science is globalized (Henrich et al., 2010; Liu et al., 2024; Sears, 1986).

Our study also reconfirmed the incomplete demographic reporting has been observed previously, i.e., age and sex are often reported but other variables such as socioeconomic status, religion, and ethnicity are rarely mentioned (Rad et al., 2018; Rozin, 2001). This persistent pattern, even in the international collaborative projects, suggests that participants' demographic information is still largely overlooked in the field. Also, our analysis revealed a lack of standardized reporting for demographics across international collaborative projects. This is understandable as the leading teams may only be able to focus on the variables of interest, and demographic information collection is subject to regulations that vary across regions. For instance, the EU's General Data Protection Regulation adds race/ethnicity data to this list (for the reidentifiability of metadata, see also de Montjoye et al., 2015). In the future, more efforts are needed to standardize the demographic reporting, especially in international collaboration (e.g., Singh et al., 2024).

Our results have important implications for interpreting the published results in psychology. First, it is important to report the demographics of participants, regardless of where the study was conducted, because the samples' representativeness has important implications for the interpretation of the results. For example, Talhelm et al. (2014) proposed an original "rice theory" to explain cultural and psychological differences within China. However, due to issues such as unrepresentative samples and other methodological concerns (Ruan et al., 2015), the generalizability of this theory remains controversial. Second, large-scale international collaborative projects should be more cautious about their claims about generalizability, because merely "having" participants from certain countries and regions does not guarantee representativeness. It is undeniable that large-scale international collaborative projects have expanded sample sizes and enhanced geographical diversity in psychological science (Liu et al., 2025), yet our results suggest that the within-country variability has largely been ignored (Ghai, 2021; Ghai et al., 2024). Third, given the current body of knowledge on human psychology and behavior is predominantly rooted in unrepresentative, young samples, highly educated participants, or WEIRD samples, conclusions derived from these samples struggle to capture the full spectrum of human psychological patterns. This is particularly

problematic for cross-cultural studies, a field dominated by WEIRD-Confucian comparisons (Krys et al., 2025). If most cross-cultural studies from China are based on young and highly educated participants in urban areas, the results may not be applicable to more than half of the population who are less educated. As a recent study also suggests, those with higher education are substantially more culturally similar to WEIRD-country populations (White & Muthukrishna, 2026).

The persistent unrepresentativeness of participants in China and other non-WEIRD regions, as well as the lack of reporting of demographics, may be attributed to two major reasons. First, neglect of representativeness is rooted in the history of psychological science. Psychological science was originated in Western countries and was then imported to China (Jing & Fu, 2001; Wang et al., 2022) and other developing countries. Researchers in developing countries adopted these methods and approaches while giving little consideration to the known flaws (e.g., the lack of representativeness) or the compatibility with local context (e.g., the huge rural-urban distinction in China). This trend is enforced by the mainstream psychological science community in the West (Syed, 2023): despite criticism of sample unrepresentativeness having been voiced since 1965 (Arnett, 2008; Henrich et al., 2010; Schultz, 1969; Smart, 1966), no substantive changes have occurred after all these years (Nielsen et al., 2017; Pollet & Saxton, 2019; Rad et al., 2018; Thalmayer et al., 2021). Even if researchers from China and other developing countries are aware of the issue of representativeness, their choices of samples are largely shaped by the cost-benefit trade-off where the cost of reaching representative samples is huge and extra pay-off is limited, as measured by number of publications or prestige of journals (see a review, de Rijcke et al., 2016; A recent real-world example is the change in submission data after eLife lost its impact factor[7]).

Second, constrained research resources further constrained researchers' efforts to reach for more representative samples globally, particularly for those in low-income countries. In most countries, funding for psychological research may not cover the cost of collecting representative samples. For researchers in low-income countries, they already grapple with

[7] The detailed information can available: https://elifesciences.org/inside-elife/c11c6101/the-elife-model-an-update-on-progress-following-changes-in-web-of-science-indexing-status

resource constraints, including insufficient funding, unstable infrastructure, time limitations, and so on (Bullock & Zúñiga, 2023; Ghai et al., 2025; Chuan-Peng et al., 2025), which further hinders their ability to conduct systematic sampling based on the study's target population. We found that 23 out of 27 projects adopted either online recruitment as the sole method or as one of multiple methods, which is easier to implement and has lower cost (Kraut et al., 2003), but also limits the participants to those who have access to and are familiar with the internet for off-line studies, convenience sampling is the dominant method for recruiting participants in the field because of its lower economic and time costs. Among the 1,000 Chinese articles we analyzed, 93.09% of those that reported participant recruitment methods used offline recruitment for participants, and the age distribution of participants peaked at 18-25, suggesting that the authors used convenience sampling. While there are methods for recruiting hard-to-rearch populations in other fields (for example, respondent driven sampling, Raifman et al., 2022), these methods are largely ignored because they are out of the comfort zone of psychological researchers and usually require extra time, training, and financial costs.

Given the current state of participant representativeness in psychological research and the above-mentioned reasons behind this lack of progress, we propose the following suggestions. First, probably the most feasible one, is the standardization of how participants' demographics are collected and reported. Comprehensive and standardized documentation of demographic backgrounds is crucial for understanding the sample representativeness, and diversity of samples, as well as the generalizability of research findings. Some promising efforts have emerged in practice. For instance, unified approach of demographic data collection tailored to young children (Singh et al., 2024), global manifesto of diverse, equitable, and inclusive open science (Ghai et al., 2025), and the Sex and Gender Equity in Research (SAGER) guidelines (Van Epps et al., 2022). Moving forward, standardized reporting guidelines for demographic information should be enforced on the journal level, ideally automatically checked by the manuscript submission system. Plenty of relevant precedents are already available, such as the Material Design Analysis Reporting (MDAR) framework in life sciences (Chambers et al., 2019) and the specific demographic reporting

requirements stipulated by journals focusing on developmental psychology (Singh et al., 2025).

Second, sample representativeness and generalizability of claims should be tested instead of assumed throughout all stages of research, not limited to the data collection stage alone (Ghai et al., 2025; Puthillam et al., 2024). Starting from the research design stage, researchers should think clearly about the target population to which the study aims to generalize. If the target population is all human beings, then the design should not only consider the sample, but also the research question itself and its protocol (materials, procedure, and tasks, etc). Before data collection, the protocol should be piloted in multiple circumstances for its validity. At the data collection stage, multiple approaches (e.g., a combination of online and offline methods) should be adopted to recruit participants, moving beyond groups that are highly educated, young, and digitally connected. When analyzing the data and reporting the results, researchers should report sample characteristics in as much detail as possible and explicitly explain the rationale for sample selection.

Third, given the practical barriers for reaching representative sample, researchers should be allowed to target a specific population, the key is being transparent and keeping intellectual humility, which actually increases trust in scientific studies (Koetke et al., 2025). Also, targeting a specific population could be relevant and of great value to one's local society. For instance, localized studies may adopt indigenous methods unfamiliar to Western researchers (Uskul et al., 2024), and such studies not only meet local needs but also hold potential value for addressing relevant issues in other cultural contexts. This requires the effort from journal editors and reviewers who allow for such "local" papers to be accepted.

Fourth, more training and tutorials are needed to lower the barriers for researchers who are interested in collecting representative samples. Currently, only a few articles have shared strategies for recruiting more representative samples, such as how to better recruit and retain individuals from lower-socioeconomic-status people (Emery et al., 2023) and how to use market-research panels to access more representative populations (Moss et al., 2023). Journals in the field should organize special issues or topics and invite experts outside psychology to facilitate the curation of resources, training materials, and best practices. The experience in

open science movement has shown that pedagogical communities are one of the most important driving forces of the adoption of open science practices (Korbmacher et al., 2023).

Finally, funding agencies should consider the financial cost for collecting representative samples if the projects are aimed at generalizing to human beings. Funding is still a major barrier for researchers, especially in low-and-middle-income countries, to adopt better scientific practices (Uskul et al., 2024; Chuan-Peng et al., 2025). To collect diverse samples, funding agencies should offer extra financial support, thereby boosting research generalizability. Extra financial support is also important to training and education on how to collect representative samples, as these methods are still new to psychological researchers (e.g., Emery et al., 2023).

Notably, we acknowledge that convenience samples can be valuable (Sherman, 2025), and even sufficient for testing theory if the theory itself is falsifiable (Dienes, 2023). And achieving representativeness across all demographic variables is practically infeasible, as sample selection is shaped by a range of factors such as research aims and available resources. Indeed, representativeness must be defined based on variables relevant to the inference target, not superficial demographic diversity (Alsalti et al., 2026). Even so, paying attention to the issue of generalizability and representativeness is largely for pragmatics reasons: many psychological studies collect data from convenience samples but claim their findings are universally generalizable (Kroupin et al., 2025).

This study has a few limitations. First, studies that were conducted in China but published in English journals were not included. Including these studies would provide a more comprehensive picture of Chinese psychological participants. Second, although our study explored the similarities and differences in patterns between participants from other countries and regions and the Chinese participants in psychological research, the actual representativeness in other countries and regions needs to be further examined by combining their national population in future research (see Liu et al., 2025).

In conclusion, we provided the first data-based profile of Chinese participants in psychological science and found that they are not representative of the Chinese population. These recurring patterns of unrepresentative samples cries for changes in actions.

Overview of the research question, hypotheses, analytical plan, interpretations of the current study, and conclusion after data analysis.

| **Question** | **Hypothesis** | **Analysis Plan** | **Interpretation given different outcomes** | **Conclusion after data analysis** |
|---|---|---|---|---|
| 1. Whether characteristics of Chinese participants reported in large-scale international collaborative projects are similar to those reported in Chinese psychological journals? | H1: There is no difference between Chinese human subjects reported in Chinese journals and in large-scale international collaborative projects. | We will visualize and compare the sample characteristics from Chinese psychological journals and international collaboration projects based on their targeted population. Also, we will use Bayesian multinomial test to inspect their similarity. | H1 is supported if the background information of the subjects in the Chinese psychological articles is similar to that in the international collaborative projects (visual inspection).<br><br>If $BF_{0a} >= 6$, we infer there is relatively strong evidence that this hypothesis is supported. | Samples from large-scale international collaborative projects and Chinese journals differed in age and geographical distribution but not in sex and educational attainment. Specifically, participants from Chinese journals were more age-concentrated, while participants from large-scale international collaborative projects showed a much larger proportion of participants from East China. Thus, H1 was partially supported. |
| 2. To what extent the Chinese participants in psychological science can represent Chinese population, as compared with the census data from the National Bureau of Statistics of China and from a large-scale social survey, Chinese Family Panel Study (CFPS). | H2: As WEIRD sample only represents a narrow slice of human beings, the Chinese samples also come from a very narrow slice of the Chinese population. | We will visualize the Chinese sample (Chinese journal articles and international collaboration projects) to inspect their representativeness relative to their targeted population from reliable data sources (e.g., census data). For example, when the articles are targeting at the general population, we will compare the sample to the whole population of census data. We will also use Bayesian multinomial test to compare the Chinese sample with census data (or CFPS or other reliable data). | H2 is supported if the subject in Chinese psychology is from a narrow slice of the entire population in China, which is estimated by the census data from the National Bureau of Statistics of China and CFPS (visual inspection).<br><br>If $BF_{a0} >= 6$, we infer there is relatively strong evidence that this hypothesis is supported. | Chinese participants have the same sex ratio as the Chinese population, but not the same in age, educational attainment and geographical distribution: Chinese participants are younger, highly educated, and from the east coast provinces. H2 was supported. |
| 3. What are the shared and distinct patterns of Chinese participants and participants from other regions? | H3: Chinese human subjects share many characteristics as most other non-WEIRD and WEIRD samples. | We will visualize the distributions of the participants from different countries and visually compare others countries with the Chinese samples. We will also compare the distributions of participants with that of the Chinese | H3 is supported if the characteristics of subjects in Chinese psychology articles share characteristics with more than half of other non-WEIRD regions or WEIRD regions in describing statistical results and visualizations. | The balanced sex pattern of Chinese participants was uncommon globally, while most countries/regions relied heavily on young adult participants. H3 was partially supported. |

| | | | | |
|---|---|---|---|---|
| | | participants, with the data from Chinese participants as expected and data from other countries as observed. | If $BF_{0a} >= 6$ for more than half of the other countries, we infer there is relatively strong evidence that this hypothesis is supported. | |

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